

Data Structures: Arrays, Lists & Trees

1. Arrays vs Linked Lists

- Arrays: $O(1)$ random access, $O(n)$ insertion/deletion, contiguous memory.
- Linked lists: $O(n)$ access, $O(1)$ insertion/deletion at a known node, no contiguity requirement.

2. Stacks & Queues

A stack is LIFO (last in, first out) — push/pop from one end. A queue is FIFO (first in, first out) — enqueue at the back, dequeue from the front.

3. Binary Search Trees

Each node has at most two children; left subtree values are smaller, right subtree values are larger.

- Search, insert, delete: $O(\log n)$ average, $O(n)$ worst case (unbalanced).
- Balanced variants (AVL, Red-Black) guarantee $O(\log n)$ worst case.

4. Complexity Cheat Sheet

- Array access: $O(1)$ | Array search: $O(n)$
- Hash table access/search/insert: $O(1)$ average
- BST balanced operations: $O(\log n)$